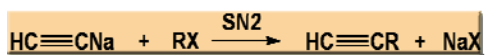
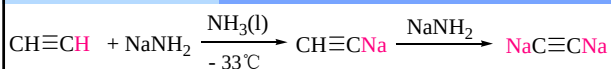
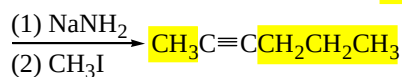
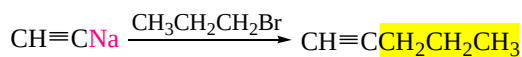


金属炔化物的生成



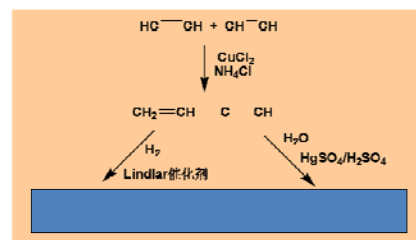
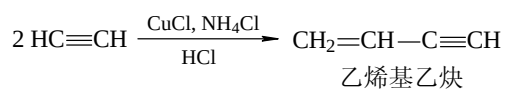
合成

熟练掌握：增长碳链的有效方法



1

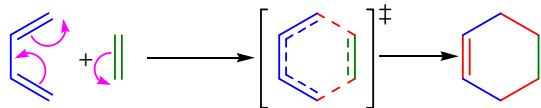
8. 乙炔的聚合 (重要)



2

双烯合成——Diels-Alder反应

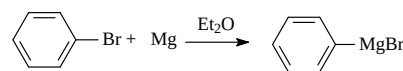
共轭二烯烃及其衍生物与含有碳碳双键、叁键等化合物进行1,4加成反应生成含六元环状的化合物



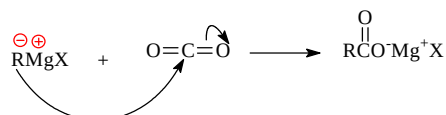
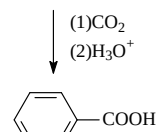
双烯体 亲双烯体

3

Grignard 试剂与二氧化碳反应

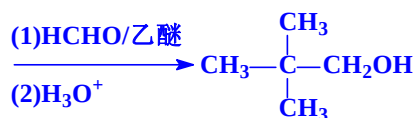
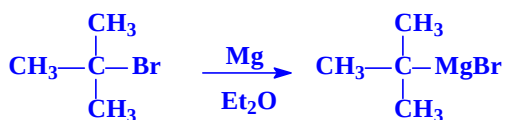


产物为增加1个碳的羧酸



4

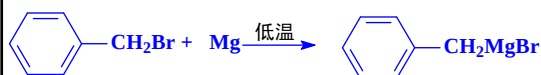
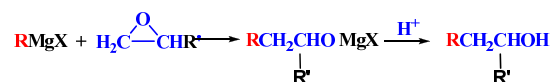
Example



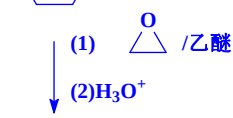
产物为增加1个碳的伯醇

5

Example

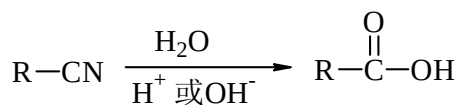


产物为增加两个碳的伯醇



6

由腈水解

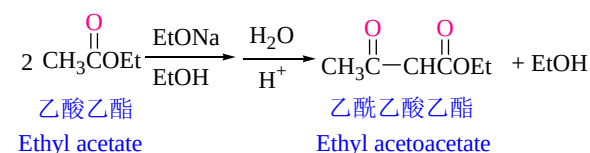


可由卤代烃制备增加一个碳原子的羧酸

7

13.4 β -酮酸酯1. β -酮酸酯的形成

Claisen 酯缩合: 含 α -氢的酯在碱 (醇钠) 的作用下, 两分子酯发生缩合反应, 生成 β -羰基酸酯同时消去一分子醇

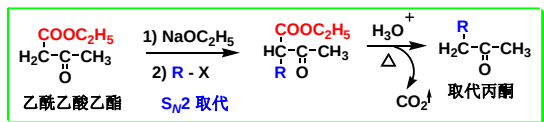


8

13.5 β -二羰基化合物的反应

1. 乙酰乙酸乙酯在合成上的应用

合成甲基酮 (掌握)

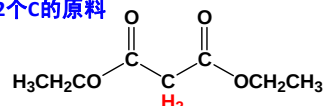
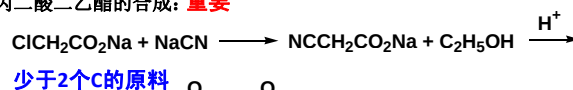


9

13.5 β -二羰基化合物的反应

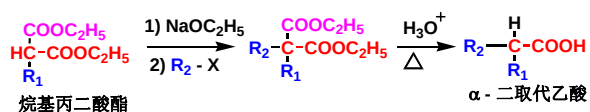
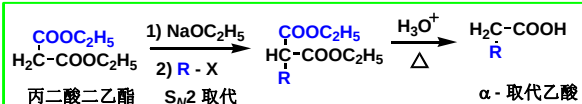
2. 丙二酸酯在合成上的应用

丙二酸二乙酯的合成: **重要**



10

合成一元羧酸

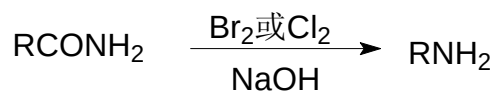


11

12.6 酰胺的特殊性质

2. Hofmann降解反应

酰胺与溴或氯在碱溶液中作用, 脱去羰基生成伯胺, 使碳链减少一个碳原子的反应, 通常称为Hofmann降解反应



12